DSA PRACTICAL 6

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AIM : Implement the Stack using link list and perform operations like INSERT, DELETE, DISPLAY

INPUT :

```
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *next;
};

struct Node *top = NULL;

void push(int item) {
    struct Node *newNode;

    newNode = (struct Node *)malloc(sizeof(struct Node));

    if (newNode == NULL) {
        printf("Stack Overflow\n");
        return;
    }

    newNode->data = item;
    newNode->next = top;

    top = newNode;
```

```
    printf("%d inserted into the stack.\n", item);
}

int pop() {
    struct Node *temp;
    int item;

    if (top == NULL) {
        printf("Stack is Empty\n");
        return -1;
    }

    item = top->data;

    temp = top;

    top = top->next;

    free(temp);

    printf("%d deleted from the stack.\n", item);

    return item;
}

void display() {
    struct Node *temp;

    if (top == NULL) {
        printf("Stack is Empty\n");
        return;
    }

    temp = top;

    printf("Stack elements are:\n");
```

```
    while (temp != NULL) {
        printf("%d\n", temp->data);
        temp = temp->next;
    }
}

int main() {
    int choice, item;

    while (1) {
        printf("\n----- STACK USING LINKED LIST -----\n");
        printf("1. INSERT (PUSH)\n");
        printf("2. DELETE (POP)\n");
        printf("3. DISPLAY\n");
        printf("4. EXIT\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter the item to insert: ");
                scanf("%d", &item);
                push(item);
                break;

            case 2:
                pop();
                break;

            case 3:
                display();
                break;

            case 4:
                printf("Program terminated.\n");
```

```
            exit(0);

        default:
            printf("Invalid choice! Please try again.\n");
      }
   }

   return 0;
}
```

OUTPUT :

```
----- STACK USING LINKED LIST -----
1. INSERT (PUSH)
2. DELETE (POP)
3. DISPLAY
4. EXIT
Enter your choice: 1
Enter the item to insert: 16
16 inserted into the stack.

----- STACK USING LINKED LIST -----
1. INSERT (PUSH)
2. DELETE (POP)
3. DISPLAY
4. EXIT
Enter your choice: 3
Stack elements are:
16

----- STACK USING LINKED LIST -----
1. INSERT (PUSH)
2. DELETE (POP)
3. DISPLAY
4. EXIT
Enter your choice: 2
16 deleted from the stack.

----- STACK USING LINKED LIST -----
1. INSERT (PUSH)
2. DELETE (POP)
3. DISPLAY
4. EXIT
Enter your choice: 4
Program terminated.


...Program finished with exit code 0
Press ENTER to exit console.
```